Tinghan Zhang

Contact Information

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Research Interests

Empirical Industrial Organization, Quantitative Marketing, (Applied) Microeconometrics

Education

Tilburg University — Tilburg, The Netherlands

Ph.D. Candidate, Econometrics, School of Economics and Management — *Sep. 2020 - Sep. 2026*

- Supervisors: Prof. dr. Tobias Klein, dr. Christoph Walsh
- Committee: Prof. dr. Bart Bronnenberg (Tilburg University), Prof. dr. Jose Moraga-Gonzalez (VU Amsterdam), dr. Johannes Kasinger (Tilburg University), dr. Ivy Dang (University of Hong Kong)

Research Master, Economics, School of Economics and Management (Cum Laude) — *Aug. 2018 - Aug. 2020*

Renmin University of China — Beijing, China

M.S., Quantitative Economics, Hanqing Advanced Institute of Finance and Economics — *Sep. 2015 - Jul. 2018*

B.A., Economics and Mathematics, School of Economics — *Sep. 2011 - Jul. 2015*

Academic Experience

Teaching Assistant, Tilburg University — 2020 - 2024

Econometrics 3, Research Master

- Panel Data, Time Series

Microeconometrics, Master

- Treatment Effects, Survival Analysis, Structural Econometrics, Non-/Semi-parametric Estimation

Research Assistant, Renmin University of China — Jan. 2016 - Jan. 2017

Working for Prof. Guangliang Ye

Visiting Researcher, Université de Genève — Sep. 2016 - Jan. 2017

Host: Prof. Harold Hau

Special Auditing Student, Kyoto University — Sep. 2014 - Feb. 2015

Host: Prof. Go Yano

Working Papers (click to view)

"**A Ranking Representation of Optimal Sequential Search**" (*Job Market Paper, in preparation for submission to Management Science*)

Sequential search models provide a powerful framework for analyzing consumer search decisions using action sequence data. Existing applications, however, typically rely on optimal rules under which subsequent decisions depend on unobserved information revealed in earlier steps. Implementing such models therefore often requires restrictive specifications and simulation-intensive procedures, increasing computational burden and limiting empirical applicability. This paper introduces a new representation of the optimal solution for a broad class of sequential search processes. We show that the outcome of an optimal sequential search process admits an equivalent representation as a partial ranking of all feasible actions. This representation enables sequential search models to be implemented through their ranking equivalents with substantial simplification. For the Weitzman-style benchmark, we develop a rank-based GHK simulator that reduces simulation requirements while improving accuracy, computational efficiency, and ease of implementation. The ranking representation further extends to a wide range of settings, including environments with partially observed action sequences and multi-stage information-acquisition processes, such as product discovery, which can be accommodated within a unified empirical approach. Overall, our results improve both the tractability and the empirical applicability of sequential search models.

"**Do I Really Want to Buy This? Preference Discovery and Consumer Search**" with Tobias Klein and Christoph Walsh *(in preparation for submission to Marketing Science, submitting invitation received from Quantitative Marketing and Economics)*

One of the most invoked assumptions in economics is that consumers know their preferences when making choices. Although theories and experiments in psychology and behavioral economics suggest that this may be unrealistic, there is relatively little evidence from the field on this question. In this paper, we use detailed clickstream data from a large Central Asian online platform to study the extent to which consumers learn about their preferences while searching for a smartphone. To quantify the speed at which this takes place and account for other factors, most notably that consumers obtain additional product information when they inspect product pages, we estimate a rich search model in which consumers learn about their willingness to pay each time they visit the checkout page. Consumers initially underestimate their price sensitivity and update it along the way. Taking this into account shows that consumers are more price-sensitive than a standard search model would predict, and that an intervention that prompts consumers to end their search early can lead to a potential welfare loss.

"**Out of Sight, Out of Cart: A Recall-based Model of Consumer Search**" (*targeting journal: Marketing Science/Management Science/Quantitative Marketing and Economics*)

We develop a structural sequential search model that incorporates imperfect recall of information acquired during search. Imperfect recall naturally arises in search environments and limits consumers' ability to make optimal purchase decisions. We identify imperfect recall using consumers' revisit actions, which allow them to reacquire decayed information, and estimate the model using rich clickstream data from a large online smartphone marketplace. Our results show that imperfect recall substantially influences both consumers' search process and purchase outcomes. Counterfactual simulations indicate that modest reductions in revisit costs, via mechanisms such as bookmarking, comparison tools, or retargeting prompts, can meaningfully reduce consumers' suboptimal purchases. These findings highlight the role of imperfect recall in consumer search and provide guidance for interventions that improve consumers' decision quality.

Selected Work in Progress

"Better Price, Better Quality? Resolve the Endogeneity in Search Decisions"

"The Causal Impact of Recommender Clicks", with Shrabastee Banerjee and George Knox

"Search Cost, Risk, and Financial Institutions Merger," with Chenxi Wang

Conferences and Seminars

2025: Barcelona Quant Marketing Workshop (as discussant, scheduled); IAAE 2026 (scheduled); 15th Consumer Search and Switching Costs Workshop (scheduled)

2025: EMAC Doctoral Colloquium; INFORMS Marketing Science Conference; 14th Consumer Search and Switching Costs Workshop; EWMES 2025; APIOC 2025*; AMA 2025*; EEA 2025*

2021-24: Tilburg Structural Econometric Lunch; Tilburg Graduate Students Society Seminar; Workshop on Digital Markets 2024

** Paper accepted but not able to present.*

Honors and Awards

Tilburg University: Koopmans Scholarships, 2018 - 2020
Renmin University of China: Excellent Student Scholarships, 2015-2017
Renmin University of China: Excellent Graduate, 2018, 2015
Kyoto University: JASSO Scholarships, 2014

Other Activities

China Banking Regulatory Commission Beijing, China
Intern in Bureau of Policy Research *Jun 2015 - Aug 2015*

Tilburg Structural Econometrics Group Tilburg, Netherlands
Co-organizer of the Lunch Seminar Series *Sep 2022 - Jun 2024*

Technical Skills

Languages: Chinese (Native), English (Proficient), Japanese (Advanced), Dutch (Beginner)
Research Software: Matlab, Julia, Stata, R
Application Software: LaTeX, Jupiter Notebook, Git

Contact Information of References

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